

## WOODRUFF LAKELAND, WI, USA

WMO: 726404

Lat: 45.928N

Lon: 89.731W

Elev: 497

StdP: 95.50

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 04865

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -25.5      | -22.4 | -29.7                         | 0.3 | -24.0 | -27.0 | 0.3 | -21.5 | 9.1                   | -6.6 | 8.3 | -6.9 | 1.1                   | 260  | 0.573 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.6                   | 29.0            | 20.5 | 27.7 | 19.7 | 26.8 | 19.1 | 22.9                | 26.6 | 21.9 | 25.7 | 21.0 | 24.3 | 4.3                  | 200  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 22.0                            | 17.7 | 25.0 | 20.9 | 16.5 | 23.8 | 19.8 | 15.4 | 23.5 | 70.5          | 26.5 | 66.5 | 25.8 | 63.2 | 24.2 | 27.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 8.5           | 7.6  | 6.9 | DB                         | -29.5 | 32.3               | 2.0 | 1.7                                                | -30.9 | 33.5       | -32.1 | 34.5       | -33.2 | 35.5       | -34.7 | 36.7 | (4) |
| (5)               |      |     | WB                         | -29.6 | 24.6               | 1.9 | 1.5                                                | -31.0 | 25.7       | -32.2 | 26.6       | -33.3 | 27.5       | -34.7 | 28.6 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 5.7         | -9.5  | -8.6  | -2.3  | 4.8   | 12.1  | 17.2  | 19.6  | 18.4  | 14.6  | 7.3   | 0.3   | -6.8  | (6)  |  |
|                                                                        | DBStd       | 11.66       | 6.73  | 6.62  | 6.86  | 5.51  | 4.89  | 3.82  | 3.27  | 3.23  | 4.68  | 5.05  | 5.51  | 6.19  | (7)  |  |
|                                                                        | HDD10.0     | 2658        | 606   | 521   | 384   | 171   | 33    | 1     | 0     | 1     | 12    | 116   | 292   | 522   | (8)  |  |
|                                                                        | HDD18.3     | 4784        | 864   | 754   | 639   | 405   | 202   | 65    | 25    | 38    | 129   | 342   | 540   | 781   | (9)  |  |
|                                                                        | CDD10.0     | 1076        | 0     | 0     | 4     | 16    | 97    | 217   | 297   | 261   | 149   | 33    | 2     | 0     | (10) |  |
|                                                                        | CDD18.3     | 161         | 0     | 0     | 0     | 1     | 8     | 31    | 63    | 41    | 16    | 1     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 1470        | 0     | 0     | 1     | 9     | 110   | 286   | 587   | 339   | 127   | 10    | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 266         | 0     | 0     | 0     | 1     | 22    | 54    | 118   | 54    | 17    | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 2.8         | 2.8   | 2.9   | 2.9   | 3.1   | 3.1   | 2.7   | 2.5   | 2.3   | 2.7   | 2.9   | 3.0   | 2.8   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 798         | 29    | 23    | 43    | 64    | 90    | 98    | 95    | 106   | 95    | 72    | 54    | 33    | (15) |  |
|                                                                        | PrecMax     | 1077        | 85    | 74    | 120   | 136   | 200   | 274   | 175   | 350   | 182   | 185   | 149   | 66    | (16) |  |
|                                                                        | PrecMin     | 408         | 3     | 0     | 6     | 8     | 9     | 35    | 28    | 10    | 13    | 5     | 11    | 4     | (17) |  |
|                                                                        | PrecStd     | 140         | 19    | 16    | 27    | 32    | 44    | 44    | 41    | 59    | 47    | 38    | 32    | 17    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 5.8   | 10.9  | 20.6  | 24.9  | 29.3  | 30.2  | 32.3  | 30.1  | 28.1  | 25.0  | 17.3  | 6.9   | (19) |  |
|                                                                        |             | MCWB        | 2.5   | 6.8   | 14.3  | 14.8  | 18.1  | 21.0  | 23.2  | 21.2  | 20.6  | 16.4  | 12.2  | 4.4   | (20) |  |
|                                                                        | 2%          | DB          | 2.9   | 6.0   | 14.1  | 21.0  | 26.5  | 27.9  | 29.1  | 27.9  | 26.8  | 22.0  | 13.9  | 4.1   | (21) |  |
|                                                                        |             | MCWB        | 1.4   | 3.4   | 7.7   | 12.2  | 17.1  | 19.6  | 21.1  | 19.8  | 19.7  | 16.2  | 10.1  | 2.6   | (22) |  |
|                                                                        | 5%          | DB          | 2.0   | 2.7   | 11.0  | 17.1  | 23.9  | 26.6  | 27.7  | 26.9  | 24.9  | 17.9  | 11.1  | 2.8   | (23) |  |
|                                                                        |             | MCWB        | 1.1   | 0.5   | 7.0   | 9.2   | 15.2  | 18.4  | 20.2  | 19.3  | 18.5  | 13.0  | 7.8   | 1.5   | (24) |  |
|                                                                        | 10%         | DB          | 0.0   | 0.2   | 6.9   | 14.1  | 22.0  | 25.0  | 26.8  | 25.1  | 22.4  | 15.2  | 7.8   | 1.2   | (25) |  |
|                                                                        |             | MCWB        | -0.7  | -1.0  | 3.8   | 7.6   | 14.7  | 17.7  | 19.6  | 18.9  | 17.6  | 11.3  | 5.3   | 0.6   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 2.8   | 7.3   | 14.7  | 15.8  | 21.1  | 23.5  | 25.2  | 24.0  | 22.9  | 19.7  | 13.4  | 5.4   | (27) |  |
|                                                                        |             | MCDB        | 4.8   | 9.1   | 18.3  | 22.0  | 25.9  | 27.2  | 27.9  | 27.2  | 26.4  | 22.2  | 16.1  | 6.1   | (28) |  |
|                                                                        | 2%          | WB          | 1.9   | 3.4   | 10.6  | 13.1  | 19.0  | 21.5  | 23.1  | 22.4  | 21.4  | 16.2  | 10.6  | 2.9   | (29) |  |
|                                                                        |             | MCDB        | 3.2   | 5.8   | 12.3  | 18.5  | 23.8  | 25.9  | 26.9  | 25.7  | 24.1  | 19.8  | 13.1  | 3.8   | (30) |  |
|                                                                        | 5%          | WB          | 0.5   | 0.6   | 6.8   | 10.5  | 17.1  | 20.4  | 22.1  | 21.4  | 20.1  | 14.2  | 7.8   | 1.8   | (31) |  |
|                                                                        |             | MCDB        | 1.6   | 2.3   | 10.3  | 15.0  | 21.3  | 24.4  | 26.1  | 24.5  | 23.4  | 18.0  | 10.2  | 2.9   | (32) |  |
|                                                                        | 10%         | WB          | -0.9  | -0.7  | 4.1   | 8.5   | 15.3  | 19.2  | 21.0  | 20.4  | 18.3  | 12.3  | 5.8   | 0.5   | (33) |  |
|                                                                        |             | MCDB        | -0.2  | 0.2   | 7.1   | 13.9  | 19.6  | 23.4  | 24.5  | 23.7  | 22.1  | 15.2  | 8.0   | 1.1   | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 8.5   | 10.0  | 10.6  | 10.9  | 12.1  | 11.4  | 11.6  | 11.2  | 10.8  | 9.0   | 7.2   | 7.2   | (35) |  |
|                                                                        |             | MCDBR       | 8.8   | 10.6  | 13.1  | 16.3  | 14.9  | 13.2  | 12.7  | 12.8  | 12.8  | 12.8  | 11.6  | 7.3   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 7.6   | 8.1   | 8.4   | 9.2   | 7.4   | 6.7   | 6.4   | 6.5   | 7.0   | 7.8   | 8.2   | 6.3   | (37) |  |
|                                                                        |             | MCWBR       | 8.3   | 9.0   | 11.4  | 14.0  | 11.7  | 11.2  | 11.0  | 10.3  | 10.2  | 10.9  | 10.3  | 6.2   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.239 | 0.264 | 0.306 | 0.330 | 0.360 | 0.359 | 0.359 | 0.358 | 0.341 | 0.305 | 0.280 | 0.249 | (40) |  |
|                                                                        |             | taud        | 2.385 | 2.310 | 2.350 | 2.436 | 2.397 | 2.419 | 2.442 | 2.450 | 2.498 | 2.582 | 2.539 | 2.429 | (41) |  |
|                                                                        | Edn at Noon | Edn at Noon | 890   | 923   | 926   | 929   | 907   | 906   | 901   | 890   | 879   | 872   | 835   | 837   | (42) |  |
|                                                                        |             | Edh at Noon | 70    | 93    | 105   | 106   | 115   | 113   | 109   | 104   | 91    | 72    | 62    | 61    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 1.44  | 2.50  | 3.73  | 4.55  | 5.37  | 5.81  | 6.03  | 5.07  | 3.91  | 2.40  | 1.40  | 1.01  | (44) |  |
|                                                                        | RadStd      | RadStd      | 0.14  | 0.26  | 0.35  | 0.46  | 0.44  | 0.45  | 0.29  | 0.33  | 0.24  | 0.26  | 0.15  | 0.11  | (45) |  |

## Historical Trends

|                              |     | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-----|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |     |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (d)                          | (e) | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)     |         |         |
| (46) Station Only            | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (36 neighbors) | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page